MEMBRANE DISC DIFFUSER

FINE BUBBLE | MODEL DN270 | 10 INCH

FEATURES

- High oxygen transfer efficiency (OTE): a large number of identical slits (pores) on the EPDM membrane produce fine air bubbles of 1 – 3 mm on average, saving power and lowering operating cost.
- Excellent backflow prevention with check valve, which avoids clogging and keeps sewage water from entering the aeration pipeline when the air supply is shut off (also available without check valve).
- EPDM membrane with high tensile strength, tear resistance and non-deformability. Its surface carries over 6500 micro pores. In operation the bubbles rise in a spiral whirlpool, giving the membrane a self-cleaning ability and a long service life.
- Unique seal ring provides tight sealing between the diffuser base and the membrane washer.
- PVC soft connector for installation on the pipeline is available as an optional accessory.



Fine Bubble Disc Diffuser · DN270

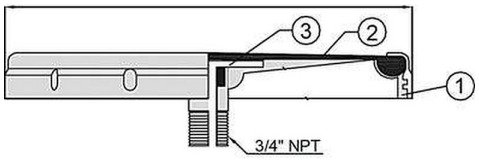
APPLICATION

- Aeration
- SBR reaction basin
- Activated sludge process
- Wastewater treatment process
- Denitrification aerobic process
- Aquafarm and other aeration applications

SPECIFICATION

PARAMETER	DN270
Disc diameter	270 mm (10 inch)
Flow rate	1 – 6 m³/h
Design flow	2.0 – 6.0 m³/h
Pressure loss	2000 – 4800 Pa
Range depth	1.5 – 7.0 m
Service area	0.5 m²
Pores quantity	6500
Bubble size	1 – 3 mm
Max. temperature	100 °C
Connector thread	3/4" NPT

CONSTRUCTION AND MATERIAL

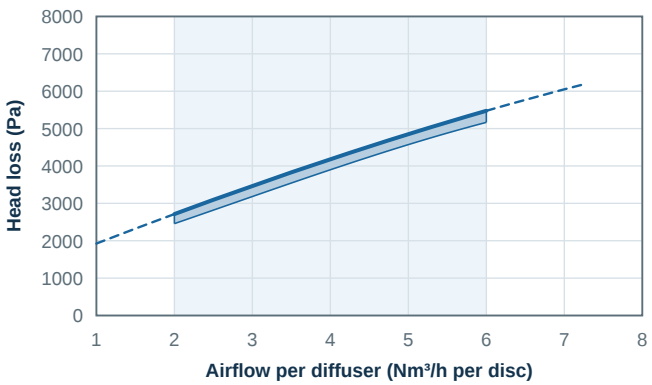


NO.	DESCRIPTION	MATERIAL
1	Retaining ring	PP
2	Membrane	EPDM
3	Disc	PP

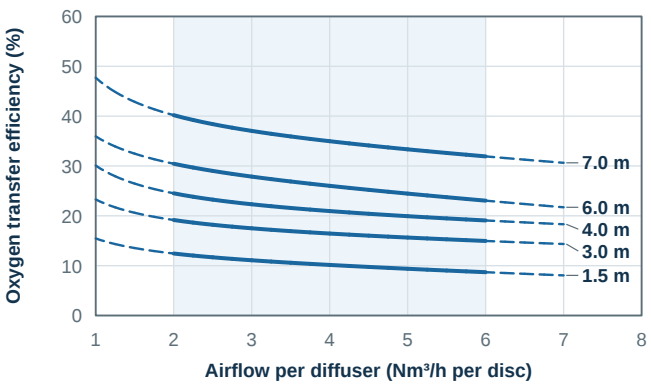
PP: polypropylene. EPDM: ethylene propylene diene monomer.

PERFORMANCE CURVE

HEAD LOSS CURVE



OTE CURVE BY WATER DEPTH



Shaded area: design flow 2.0 – 6.0 m³/h. Dashed line: outside the design flow.

perotac
MADE IN TAIWAN

MEMBRANE DISC DIFFUSER

COARSE BUBBLE | MODEL DN128 | 5 INCH

FEATURES

- Special structure prevents water backflow during air flow interruptions.
- Main body made of high-quality ABS with heat resistance; resists softening at high temperature.
- The membrane offers durable protection against corrosion, weather, oil and deformation.
- Adjustable discharge holes for different air volumes reduce pressure loss, prevent clogging and keep maintenance low.
- Easy installation; can be installed facing upwards or downwards.

APPLICATION

- Aeration
- Wastewater treatment
- Activated sludge process
- Water purification process
- Mixing
- Equalization process

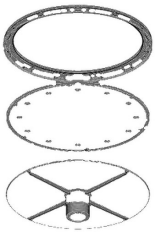
SPECIFICATION

PARAMETER	DN128
Disc diameter	**128 mm (5 inch)**
Rubber cap	**EPDM**
Main body	**ABS**
Connector thread	**3/4" NPT (male)**
Flow rate	**6 – 24 m³/h**
OTE	**17 % @ water depth 4 m**
Head loss	**1550 Pa @ 15 m³/h**



Coarse Bubble Disc Diffuser · DN128

CONSTRUCTION AND MATERIAL

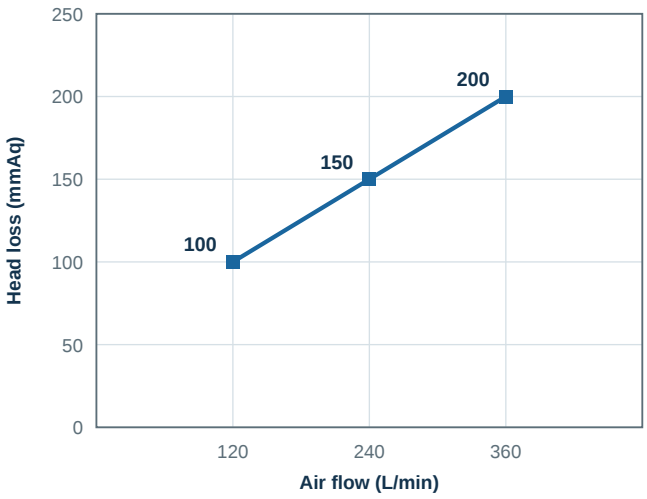


NO.	DESCRIPTION	MATERIAL
1	Rubber cap	**EPDM**
2	Main body	**ABS**

EPDM: ethylene propylene diene monomer. ABS: acrylonitrile butadiene styrene.

PERFORMANCE CURVE

HEAD LOSS CURVE



OTE CURVE

